

JomCode Coding Bootcamp

Values, types and operators

Understand, predict, type, run, then practice.

What you will be able to do

- 01 Identify int, float, str, bool and None values.
- 02 Use arithmetic and assignment to model workshop data.
- 03 Predict precedence and spot type mismatches.

Plain meaning → technical view → use

PLAIN MEANING

Programs work with numbers, text, yes/no values and missing values.

TECHNICAL VIEW

Python int, float, str, bool and None have different operations and semantics.

WHEN YOU USE IT

Capacity and registration counts must remain numbers.

TIP

Run ``python chapter_03_values_types_and_operators.py``; predict the type before printing it.

Worked example 1: Types in a workshop

`chapter_03_values_types_and_operators.py`

```
1 capacity = 20
2 price = 12.5
3 title = "Python Basics"
4 is_open = True
5 cancelled_at = None
6 print(type(capacity).__name__)
7 print(type(price).__name__)
8 print(type(title).__name__)
9 print(is_open, cancelled_at)
```

PREDICT Which values can count registrations?

Example 1: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
int
float
str
True None
```

WHY THIS HAPPENS

Use an integer for a count. Boolean is a separate intent even though Python treats bool as an int subclass.

Exercise 1: Type diagnosis

BUILD

Fix a total that concatenates numeric-looking text. Convert both known valid inputs explicitly.

CHECK

"12" and "8" => 20. Explain why "12" + "8" produces "128".

IF STUCK

int accepts valid integer text. Invalid-input handling arrives later.

Worked example 2: Seat arithmetic



chapter_03_values_types_and_operators.py

```
1 capacity = 18
2 registered = 7
3 remaining = capacity - registered
4 print(remaining)
5 print(18 / 4)
6 print(18 // 4)
7 print(18 % 4)
```

PREDICT How many complete groups of four fit?

Example 2: output and reason

TERMINAL	>_	EXPECTED OUTPUT
----------	----	-----------------

11		
4.5		
4		
2		

WHY THIS HAPPENS

Floor division counts complete groups. Modulo reports leftover seats.

Exercise 2: Groups and leftovers

BUILD

For 23 attendees in groups of 5, compute complete groups and leftovers.

CHECK

23 => 4 groups and 3 leftover. 0 => 0 and 0. 25 => 5 and 0.

IF STUCK

Use // and % with the same divisor.

Worked example 3: Precedence and reassignment



chapter_03_values_types_and_operators.py

```
1 total = 2 + 3 * 4
2 print(total)
3 total = (2 + 3) * 4
4 print(total)
5 registered = 4
6 registered += 1
7 print(registered)
```

PREDICT Does += create a second registration variable?

Example 3: output and reason

TERMINAL	>_	EXPECTED OUTPUT
----------	----	-----------------

14		
20		
5		

WHY THIS HAPPENS

For immutable numbers it computes a new value and rebinds the same name.

Exercise 3: Remaining seats

BUILD

Store capacity 24 and registered 9. Print remaining seats with descriptive variable names.

CHECK

24,9 => 15. 10,10 => 0. No string arithmetic.

IF STUCK

Subtraction belongs between integer values.

Checkpoint

PROVE IT

Explain the types and arithmetic behind remaining seats without looking at a solution.